Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject: Artificial Intelligence (01CT0616)	**Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression**	
Experiment No: 02	**Date:**	**Enrollment No: 92410133020**

Aim: To obtain the best fit line over single feature scattered datapoints using Linear Regression

IDE: Google Colab

Theory:

Linear regression is a method for determining the best linear relationship between two variables *X* and *Y*. If variables *X* and *Y* are uncorrelated, it is pointless embarking upon linear regression. However, if a reasonable degree of correlation exists between *X* and *Y* then linear regression may be a useful means to describe the relationship between the two variables. The usual approach is to use the *least-squares* method, which minimizes the squared difference between the actual data points and a straight line. Let $[x_i, y_i]$, $i = 1,2,3,\ldots,N$ be the *N* pairs of data values of the variables *X* and *Y*. The straight-line relating *X* and *Y* is $y = mx + c$, where *m* and *c* are the gradient and constant values (to be determined) defining the straight line. Thus, $y(x_i) - y_i$ is the difference between the line and data point *i* (see Fig. 1). Taking all the data points, we seek values of *m* and *c* that minimize the squared difference *SD*.

$$\sum_{1}^{N}\left[y(x_i) - y_i\right]^2$$

This is achieved by calculating the partial derivatives of *SD* with respect to *m* and *c* and finding the pair [*m*,*c*] such that *SD* is at a minimum.

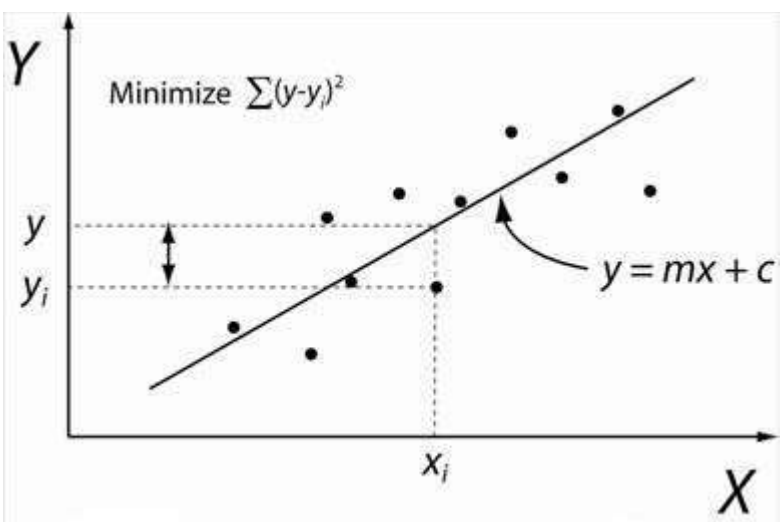

Figure 1: Illustration of Linear Regression. Linear least squares regression, the idea is to find the line y = mx + c that minimizes the mean squared difference between the line and the data points

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Batch Gradient Descent:

Gradient Descent is an optimization algorithm used for minimizing the cost function in various machine learning algorithms. It is basically used for updating the parameters of the learning model. Batch gradient descent which processes all the training examples for each iteration of gradient descent. But if the number of training examples is large, then batch gradient descent is computationally very expensive.

Let m be the number of training examples. Let n be the number of features.

Algorithm for batch gradient descent :

Let $h_\theta(x)$ be the hypothesis for linear regression. Then, the cost function is given by:
Let Σ represents the sum of all training examples from i=1 to m. $J_{train}(\theta) = (1/2m)\ \Sigma(h_\theta(x^{(i)}) - y^{(i)})^2$

Repeat {
$\theta j = \theta j - (\text{learning rate}/m) * \Sigma(h_\theta(x^{(i)}) - y^{(i)})x_j^{(i)}$
For every j =0 …n
}
Where $x_j^{(i)}$ Represents the jth feature of the ith training example. So if *m* is very large(e.g. 5 million training samples), then it takes hours or even days to converge to the global minimum.That's why for large datasets, it is not recommended to use batch gradient descent as it slows down the learning.

Pre Lab Exercise:

a. Explain the meaning of linear regression

sol:Linear Regression is a supervised machine learning algorithm used to find the relationship between an independent variable (X) and a dependent variable (Y). It fits a straight line of the form:

$$y=mx+c$$

where:

- m = slope (rate of change)
- c = intercept

The goal is to minimize the error between predicted and actual values using methods like least squares.

b. Write three applications of linear regression

- House Price Prediction – Predict price based on area, rooms, location
- Sales Forecasting – Estimate future sales based on past data
- Student Performance Prediction – Predict marks based on study hours

c. Write three advantages of linear regression

- Simple and easy to understand
- Fast computation and efficient
- Works well with small datasets

d. Write three limitations of linear regression

- Assumes linear relationship (not suitable for complex data)
- Sensitive to outliers
- Cannot capture non-linear patterns

Methodology:

1. Load the basic libraries and packages
2. Load the dataset
3. Analyse the dataset
4. Pre-process the data
5. Visualize the Data
6. Separate the feature and prediction value columns
7. Write the Hypothesis Function
8. Write the Cost Function
9. Write the Gradient Descent optimization algorithm
10. Apply the training over the dataset to minimize the loss
11. Find the best fit line to the given dataset
12. Observe the cost function vs iterations learning curve

Program (Code):

To be attached with

```
import numpy as np
import matplotlib.pyplot as plt

X = np.array([1, 2, 3, 4, 5, 6])
Y = np.array([2, 3, 5, 7, 8, 9])
```

```
plt.scatter(X, Y)
plt.title("Scatter Plot of Data Points")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()

m = 0
c = 0
learning_rate = 0.01
iterations = 1000
n = len(X)

cost_history = []

for i in range(iterations):
    dm = 0
    dc = 0

    for j in range(n):
        y_pred = m * X[j] + c
        dm += -(2/n) * X[j] * (Y[j] - y_pred)
        dc += -(2/n) * (Y[j] - y_pred)

    m = m - learning_rate * dm
    c = c - learning_rate * dc

    cost = np.sum((Y - (m*X + c))**2) / (2*n)
    cost_history.append(cost)

print(f"y = {m:.3f}x + {c:.3f}")

Y_pred = m * X + c

plt.scatter(X, Y)
plt.plot(X, Y_pred)
plt.title("Best Fit Line")
plt.xlabel("X")
plt.ylabel("Y")
plt.show()

plt.plot(range(iterations), cost_history)
plt.title("Cost vs Iterations")
plt.xlabel("Iterations")
plt.ylabel("Cost")
plt.show()
```

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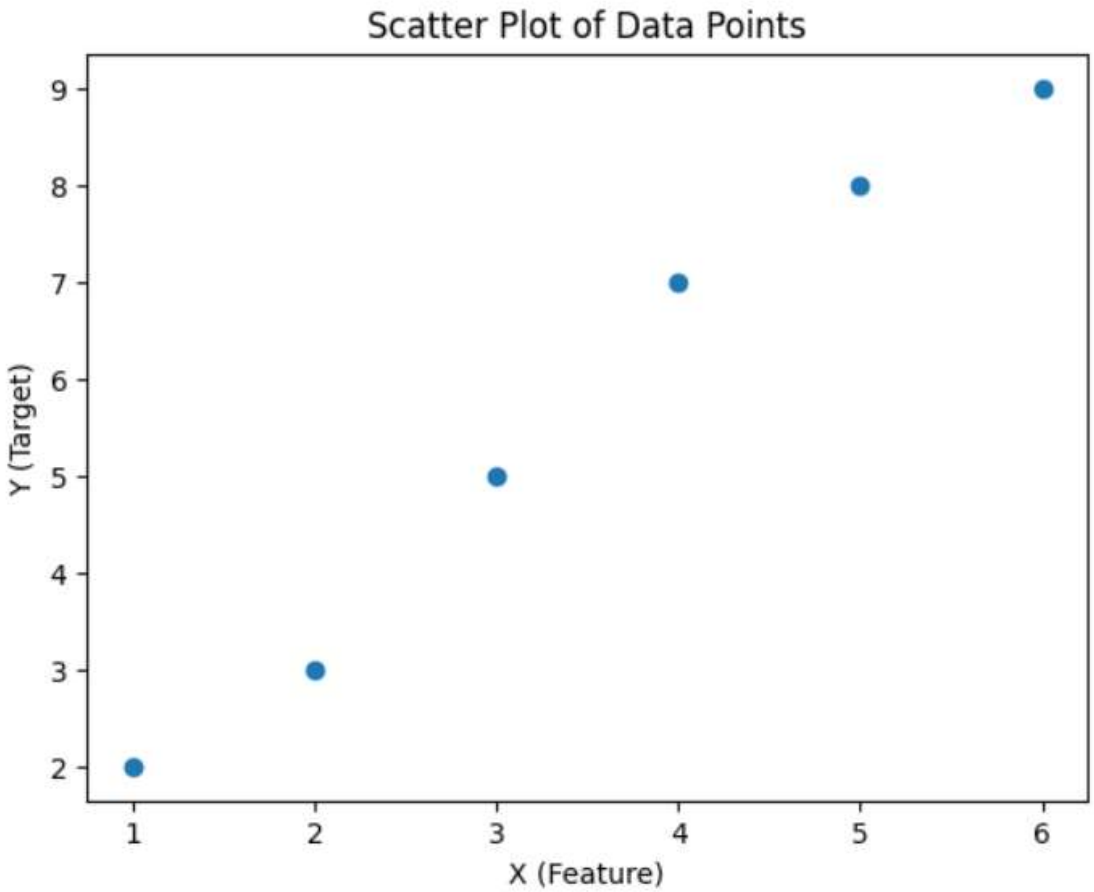
Results:

To be attached with

a. Datapoints scattering (without best fit line)
b. Scatter plot showing the Best fit line in the last iteration of training



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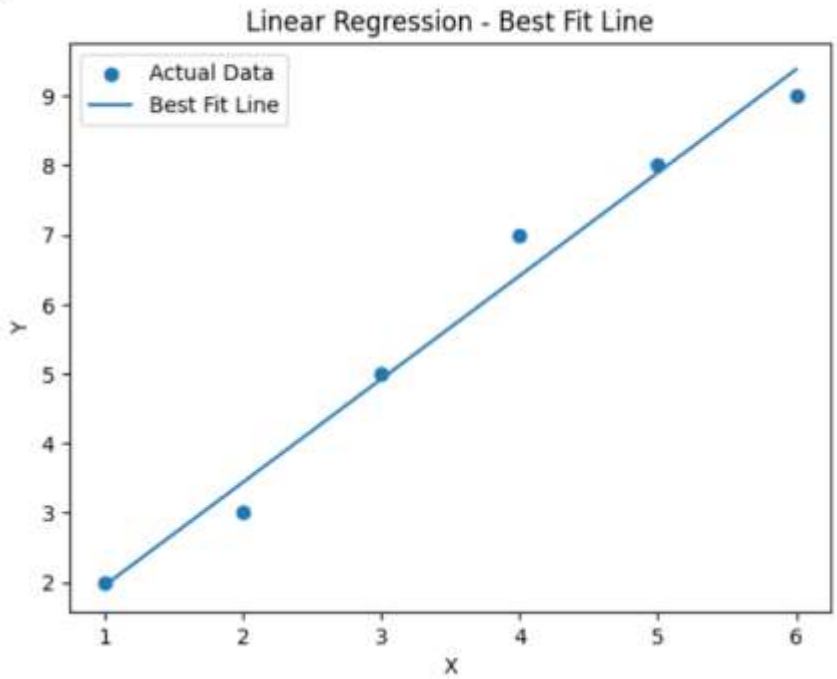
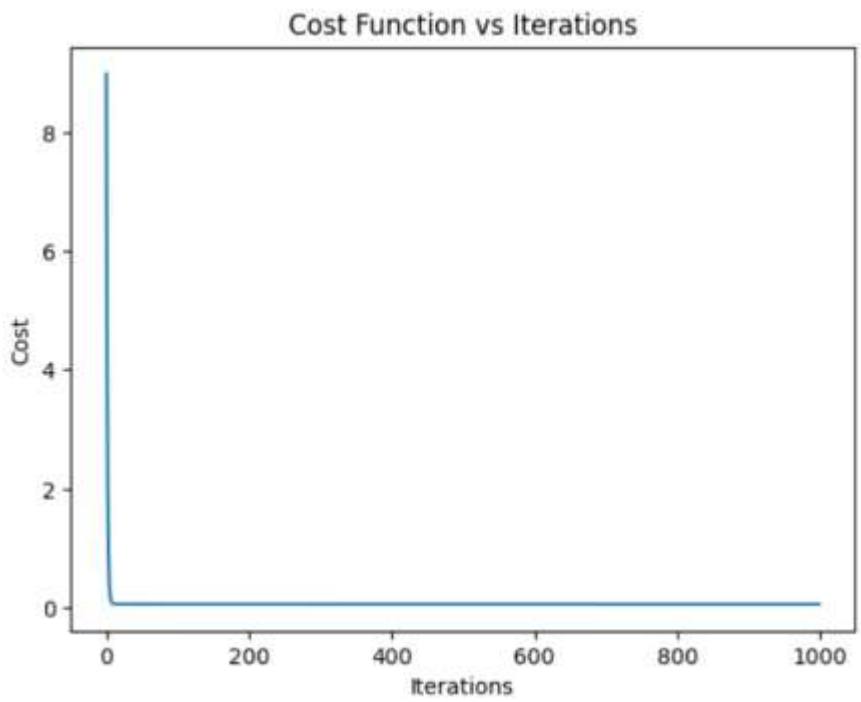
```
Best Fit Line:
y = 1.486x + 0.464
```





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Observation and Result Analysis:

a. Nature of the dataset
   - Dataset shows a linear relationship between input and output
   - Data points are scattered around a straight line
   - Some variation/noise may exist

b. During Training Process
   - Cost function gradually decreases
   - Model parameters (m and c) are updated using gradient descent
   - Learning improves with each iteration
   - Convergence depends on learning rate

c. After the training Process
   - Best fit line is obtained
   - Minimum error achieved
   - Predictions closely match actual values
   - Model becomes stable

Post Lab Exercise:

a. What are the major assumptions considered in linear regression
   - Linear relationship between X and Y
   - Independence of errors
   - Constant variance (Homoscedasticity)
   - No multicollinearity
   - Errors are normally distributed

b. Why MSE is used instead of MAE for calculating the loss function

   MSE (Mean Squared Error):
   - Penalizes large errors more (due to squaring)
   - Differentiable → easy for gradient descent

   MAE:
   - Not smooth (harder for optimization)

Hence, MSE is preferred for optimization.

c. How can the behaviour of outliers be understood while dealing with the unseen dataset

- Outliers increase error significantly
- They can distort the regression line
- Can be handled using:
  - Data cleaning
  - Robust regression
  - Visualization (scatter plots)

d. Derive the Normal Equation for the Linear Regression.

We want to minimize:

$$J(\theta)= 2m\ 1\ \ (X\theta-y)\ T\ (X\theta-y)$$

Take derivative and set to zero:

$$\partial\theta\ /\partial J\ =0$$

$$X\ T\ X\theta=X\ T\ y$$

Final solution:

$$\theta=(X\ T\ X)\ -1\ X\ T\ y$$